

Optimizing Prompting Strategies Improves Large Language Model Classification of Pain- and Fatigue-Related Functional Impact in Childhood Cancer Survivors

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

****Dear Authors and Editors,****

Thank you for the opportunity to review the manuscript titled “Optimizing Prompting Strategies Improves Language Model Classification of Symptom Narratives in Childhood Cancer Survivors.” In it, the authors evaluate how prompting strategies affect the performance of ChatGPT-4o in classifying whether pain or fatigue symptoms impact are impacting physical, social, or cognitive functioning.

This is an important and timely question. However, I found the paper somewhat too narrow to fully address it. The exclusive use of a single closed-weight commercial model (ChatGPT-4o) limits both the reproducibility and broader applicability of the findings. In addition, the manuscript misses the opportunity to directly compare results against the authors’ prior work using predictive NLP models, which could substantially strengthen the contribution of this study.

**Strengths**

1. Clear, well-written manuscript
2. Interesting and clinically relevant research question
3. Use of appropriate evaluation metrics
4. Well-designed and accessible figures
5. I appreciate that the introduction that acknowledges the value of both LLM and non-LLM NLP approaches - something often overlooked in papers I review

**Major Comments**

1. ****Exclusive use of GPT-4o****

The study’s main limitation is the reliance on a single model. Since the stated aim (paper’s title) is to understand how prompting strategies influence symptom classification, restricting the analysis to one foundation model significantly limits the generalizability of the conclusions. Findings may be specific to ChatGPT-4o - which changes over time and is already out-of-date with ChatGPT 5’s release - rather than to LLMs more broadly.

I also question the description of ChatGPT-4o as a “widely pragmatic, accessible LLM.” In practice, this is a closed-weight, commercial system with usage restrictions.

For reproducibility and scientific impact, I strongly encourage the authors to repeat the analyses using at least one open-weight model (e.g., from the LLaMA family or other widely available families). Ideally, including models from two different creators would allow readers to see whether prompting strategies yield consistent effects across platforms. This would considerably strengthen the study’s value and understand if the author’s findings are generalizable.

2. **Comparison against predictive NLP models**

Reading the paper, I spent much of it wondering "well why not compare against trained predictive models". Only late in the discussion is it noted that the authors have already tested predictive models in prior work (besides a vague citation in the Introduction). This connection should be made clear in the Introduction, with an explicit statement that the current study builds on that earlier work.

More importantly, I recommend including the prior results in a comparative table alongside the current findings. This would allow readers to directly assess whether advanced prompting strategies bring LLMs closer to the performance of trained predictive models. Such a comparison is rare in the literature and would make this study's contribution much stronger - especially if the best prompting approaches approximate the performance of fine-tuned BERT models.

Minor Comments

1. Please clarify the data split. The manuscript states that interviews were split into train/test sets - does this mean the split was at the level of patients? If the same patient contributed meaning units to both training and test sets, that would be an important limitation.

2. The sentence "This study evaluated how a large language model, ChatGPT-4o, classifies physical, social, and cognitive functional aspects of pain and fatigue symptoms using four prompting strategies: zero shot, few-shot, Chain-of-Thought, and Generated Knowledge" could be simplified for clarity. For example: "This study evaluated how ChatGPT-4o, using four prompting strategies (zero-shot, few-shot, Chain-of-Thought, and Generated Knowledge), classified whether pain or fatigue symptoms affected physical, social, or cognitive functioning."

3. I somewhat feel the title could be tweaked regarding "classification of symptom narratives" I was expecting the paper to be around identifying whether symptoms were present, instead of classifying how symptoms are affecting a patient. Also, the authors should add "large" as they are not looking at language models generally here.

Reviewer #2

(Remarks to the Author)

Thank you for the opportunity to read through this well-written and interesting paper. The manuscript explores the use of advanced NLP methods—particularly LLMs (ChatGPT)—for symptom interpretation in childhood cancer survivorship. The study applies zero-shot, few-shot, and Chain-of-Thought prompting approaches to model semantic representations of pain and fatigue. The integration of structured frameworks (PROMIS) and fairness considerations supports both methodological rigor and clinical sensitivity. Overall, the work is timely and relevant, offering novel insights into the intersection of AI, cancer care, and patient-reported outcomes. However, a few clarifications, definitional refinements, and structural adjustments are needed to strengthen the paper.

1. The manuscript introduces concepts such as bidirectional contextual representations of text, zero-shot prompting, few-shot prompting, and Chain-of-Thought prompting. While these are well described in the Methods section, they could be first presented in the Introduction to enhance readability through the paper as structured (Introduction>Results>Discussion>Methods).

2. Please ensure acronyms are defined on first use. For example, I think there is an issues with "Chain-of-Thought" and "Chain-of-Thought (COT)".

3. Terms such as accuracy (0.740, 95% CI: 0.658–0.809), precision (0.712, 95% CI: 0.577–0.817), recall, and F1-score require clearer interpretation in the context of symptom classification. Specifically, what constitutes "good" precision for these models? Given the clinical implications of false positives versus false negatives, some discussion is warranted.

4. The manuscript also references area under the receiver operating characteristic curve (AUROC) and area under the precision-recall curve (AUPRC). These are important but may require explanation for a mixed clinical/technical readership.

5. It remains unclear why pain and fatigue were prioritized over other dimensions of symptom experience.

6. Relatedly, the terms "social attribute" and "cognitive attribute" are not common in clinical/symptom science vernacular. Alternative phrasing should be considered to align more closely with established clinical terminology.

7. The sentence "The integration of structured frameworks, such as the Patient-Reported Outcomes Measurement Information System (PROMIS) scales, further supported the clinical relevance of COT-guided interpretation" is somewhat ambiguous. Please clarify the connection between PROMIS, clinical relevance, and the COT interpretations.

8. The discussion would benefit from more explicit commentary on practical applications. For instance: How might clinicians incorporate AI-driven interpretation of survivor-reported text into routine practice? What safeguards, training, or decision-support frameworks would be required for integration into clinical workflows?

9. The dataset consisted of 395 semantic units for pain and 424 for fatigue. The adequacy of this dataset size requires explicit discussion. While qualitative richness may be high, quantitative reliability for training/testing modern LLMs is less certain. Please provide justification or acknowledge limitations.

10. In addition, the sentence reporting these dataset counts seems misplaced in its current location. I recommend moving it to the Data Collection section for clarity.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I would like to thank the authors for their thorough and thoughtful revisions. In particular, I appreciate the addition of the open-weight model comparison (Llama-3.1-8B-Instruct) alongside GPT-4o, and the inclusion of prior BERT results in the comparative performance tables. These additions substantially strengthen the manuscript by improving transparency, reproducibility, and generalizability.

I also thank the authors for engaging with my concern regarding potential information leakage. In the rebuttal, they clearly explain that the dataset was split at the meaning-unit level within each interview/participant (70% learning and 30% testing), and they appropriately acknowledge that within-participant clustering may inflate performance estimates. I further appreciate the clarification that the LLM-based/prompting approaches were not “trained” in the traditional sense.

However, I do not believe this limitation is yet sufficiently acknowledged in the revised manuscript itself. Even without model training, the approach still involves prompt engineering and procedure development on the learning subset; therefore, a within-participant split may still yield optimistic performance estimates due to lexical/stylistic dependence between learning and test meaning units from the same participant. This also limits conclusions about generalization to unseen patients.

I therefore request that the authors explicitly add this point to the Limitations section and note that future work should assess robustness using participant-level separation (e.g., patient-level train/test split). Assuming this is added, I have no further comments.

Reviewer #2

(Remarks to the Author)

I thank the authors for their careful considerations of my suggestions. I am quite satisfied with the associated changes made and have no other recommendations. I appreciated the chance to engage with this work.

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COMMSMED-25-1565

Revised title: Optimizing Prompting Strategies Improves Large Language Model Classification of Pain- and Fatigue-Related Functional Impact in Childhood Cancer Survivors

Old title: Optimizing Prompting Strategies Improves Language Model Classification of Symptom Narratives in Childhood Cancer Survivors

Overall Response:

Thank you for the opportunity to revise our manuscript. In the response letter below, two reviewers' comments are italicized, and our point-by-point response to each comment is written in regular font. In the revised manuscript, we used underlined, red-colored texts to indicate where changes have been made.

Reviewer #1:

Thank you for the opportunity to review the manuscript titled “Optimizing Prompting Strategies Improves Language Model Classification of Symptom Narratives in Childhood Cancer Survivors.” In it, the authors evaluate how prompting strategies affect the performance of ChatGPT-4o in classifying whether pain or fatigue symptoms impact are impacting physical, social, or cognitive functioning.

This is an important and timely question. However, I found the paper somewhat too narrow to fully address it. The exclusive use of a single closed-weight commercial model (ChatGPT-4o) limits both the reproducibility and broader applicability of the findings. In addition, the manuscript misses the opportunity to directly compare results against the authors' prior work using predictive NLP models, which could substantially strengthen the contribution of this study.

Response:

We thank the reviewer for this important and constructive comment. In response to this overall comment, we have expanded the scope of the study into two key aspects. First, to address concerns regarding reproducibility and reliance on a single closed-weight model, we extended our analyses to include an open-weight large language model (Meta's Llama), as detailed in our responses to this reviewer's Comments #1, 2, and 3. Second, to strengthen contextualization with prior work, we revised current Tables 3 and 4 to enable direct performance comparisons between the traditional predictive NLP model (BERT) from our prior publication and the large language models (ChatGPT and Llama) evaluated in the current study. This addition is also described in our response to this reviewer's Comment #5. Together, these revisions broaden the methodological scope, enhance reproducibility, and allow readers to directly compare model performance across NLP and LLM paradigms.

Major Comments

Exclusive use of GPT-4o

1. The study's main limitation is the reliance on a single model. Since the stated aim (paper's title) is to understand how prompting strategies influence symptom classification, restricting the analysis to one foundation model significantly limits the generalizability of the conclusions.

Findings may be specific to ChatGPT-4o - which changes over time and is already out-of-date with ChatGPT 5's release - rather than to LLMs more broadly.

Response:

Thank you for raising this important point regarding model dependence and generalizability. We agree that reliance on a single foundation model can limit the broader applicability of findings, and we therefore took deliberate steps to address this concern in the revised manuscript.

First, we selected ChatGPT-4o because all experiments were completed in September 2024, when it was the most stable, widely deployed, and extensively benchmarked large language model, with documented evaluations across clinical and biomedical reasoning tasks. ChatGPT-5 was released only after the data lock and completion of all analyses. Substituting a newer model post hoc would compromise reproducibility by confounding the effects of the prompting strategy with performance gains attributable to model evolution, including architectural scaling and enhanced built-in reasoning. Because the central objective of this study is to isolate the effect of prompting strategies on symptom classification, introducing a substantially more capable model would preclude disentangling prompt-level effects from improvements driven by the model itself.

Second, to directly address generalizability beyond a single proprietary system, we extended the full experimental pipeline to an open-weight foundation model, Meta Llama-3.1-8B-Instruct. This model was selected because it is stable, fully documented, and reproducible, enabling transparent cross-model comparison. As with GPT-5, we did not include rapidly evolving frontier models (e.g., Llama-4) because their ongoing architectural changes pose similar risks of prompt-model confounding. The highly consistent prompting effects observed across ChatGPT-4o and Llama-3.1-8B-Instruct, despite their distinct architectures and training paradigms, indicate that our findings reflect robust prompt design effects rather than model-specific behavior. This cross-model replication strengthens the generalizability of our conclusions to large language models more broadly.

2. I also question the description of ChatGPT-4o as a “widely pragmatic, accessible LLM.” In practice, this is a closed-weight, commercial system with usage restrictions.

Response:

We appreciate the reviewer’s clarification and agree that our original wording overstated the accessibility of ChatGPT-4o. We have revised the manuscript throughout to accurately describe ChatGPT-4o as a commercial, closed-weight large language model with usage restrictions, rather than as an open or broadly accessible system. To clearly reflect this distinction and strengthen conceptual clarity, we added the following sentence to the sixth paragraph of the **Introduction** section: “While prior studies have focused on BERT- or ChatGPT-based models for symptom research,^{14,18,22,25,27} to our knowledge, none have systematically examined whether symptom-classification performance generalizes across LLM families with distinct architectures. ChatGPT-4o and Llama-3.1 represent two leading but different model paradigms: the former is a proprietary, closed-weight model optimized through extensive instruction tuning, whereas the latter is an open-weight model designed for reproducible and flexible deployment. These architectural and accessibility differences may affect how models interpret cancer survivors’ narrative descriptions of physical, social, and cognitive impact of symptoms. Evaluating both platforms allows assessment of whether prompting effects are robust across LLM ecosystems rather than specific to a single system.”

3. For reproducibility and scientific impact, I strongly encourage the authors to repeat the analyses using at least one open-weight model (e.g., from the LLaMA family or other widely available families). Ideally, including models from two different creators would allow readers to see whether prompting strategies yield consistent effects across platforms. This would considerably strengthen the study's value and understand if the author's findings are generalizable.

Response:

We agree that open-weight replication is essential for reproducibility and generalizability. In direct response, we repeated the full experimental pipeline using Llama-3.1-8B-Instruct, a fully documented open-weight model developed by Meta, representing a distinct creator and model family from OpenAI's ChatGPT-4o. All prompts, preprocessing, and evaluation procedures were held constant to isolate prompt effects from platform-specific effects. The highly consistent prompting patterns observed across both models demonstrate that our findings reflect prompt strategy rather than proprietary model behavior, substantially strengthening scientific rigor and cross-platform generalizability. We did not include rapidly evolving frontier models (e.g., GPT-5, Llama-4) due to frequent model updates, which create substantial reproducibility challenges (see our response to Reviewer #1's first question).

Comparison against predictive NLP models

4. Reading the paper, I spent much of it wondering "well why not compare against trained predictive models". Only late in the discussion is it noted that the authors have already tested predictive models in prior work (besides a vague citation in the Introduction). This connection should be made clear in the Introduction, with an explicit statement that the current study builds on that earlier work.

Response:

We thank the reviewer for this excellent suggestion. We have now made this connection explicit and prominent in the third paragraph of the **Introduction** section (after discussing NLP and LLM approaches), which reads: "Our team previously developed and validated fine-tuned BERT models for classifying pain and fatigue from symptom narratives in the St. Jude Lifetime Cohort Study.¹⁴ While these models achieved strong performance, they required substantial computational resources, specialized NLP expertise, and large volumes of labeled training data. Moreover, their dependence on task-specific architectures and extensive annotation limits flexibility and scalability for broader clinical deployment."

5. More importantly, I recommend including the prior results in a comparative table alongside the current findings. This would allow readers to directly assess whether advanced prompting strategies bring LLMs closer to the performance of trained predictive models. Such a comparison is rare in the literature and would make this study's contribution much stronger – especially if the best prompting approaches approximate the performance of fine-tuned BERT models.

Response:

Thank you for this suggestion that substantially strengthens our manuscript. We have revised Tables 3 and 4 to present side-by-side performance metrics for BERT (from our prior work) and for the four prompting strategies applied to ChatGPT-4o and Llama-3.1 in the present study.

In addition, we now discuss these performance differences in the fourth paragraph of the **Discussion** section, which reads: “Direct comparison with our previously fine-tuned BERT models,¹⁴ highlights key trade-offs between supervised and prompt-based approaches. Across both pain and fatigue domains, BERT showed the most stable and consistently strong performance, particularly for physical and cognitive functional impact, reflecting the advantages of task-specific fine-tuning on labeled data. Although ChatGPT-4o achieved slightly lower overall performance, COT and GK prompting enabled competitive results, especially for fatigue-related physical and cognitive impact and, in select cases, social impact. These patterns reflect fundamental modeling differences: BERT is optimized for supervised classification using domain-labeled data,¹⁷ whereas LLMs rely on prompt-driven inference without task-specific parameter updates. From a translational perspective, LLMs reduce dependence on large, annotated datasets by shifting effort toward prompt design and domain expertise, thereby facilitating deployment in settings without machine-learning infrastructure, whereas BERT may be preferred when fine-tuning resources and labeled data are available.”

Minor Comments

1. Please clarify the data split. The manuscript states that interviews were split into train/test sets – does this mean the split was at the level of patients? If the same patient contributed meaning units to both training and test sets, that would be an important limitation.

Response:

Thank you for this constructive question. We clarify that the train-test split in this study was performed at the level of meaning units rather than at the level of study participants. Specifically, within each interview, 70% of extracted meaning units were used for model development and 30% for testing, as described in the Model Learning and Evaluation subsection under the **Methods** section. This design choice reflects the study’s analytic focus, in which the primary unit of analysis is the individual symptom statement rather than the patient. Each meaning unit represents a short, self-contained symptom description and is independently assigned a functional impact label by expert reviewers. Importantly, neither ChatGPT-4o nor Llama-3.1 was trained or fine-tuned on the study data, and neither model updates internal parameters during evaluation; each prediction is generated independently for each symptom statement. As such, model performance reflects the ability to classify previously unseen symptom narratives rather than to learn or exploit participant-specific patterns. We acknowledge that clustering of meaning units within participants may introduce residual stylistic similarity, which could modestly inflate model performance estimates. However, this less likely jeopardizes the internal validity of comparative analyses, as all prompting strategies and models were evaluated under an identical data-splitting framework. Future large-scale validation studies can incorporate participant-level separation to assess cross-patient generalizability.

2. The sentence “This study evaluated how a large language model, ChatGPT-4o, classifies physical, social, and cognitive functional aspects of pain and fatigue symptoms using four prompting strategies: zero shot, few-shot, Chain-of-Thought, and Generated Knowledge” could be simplified for clarity. For example: “This study evaluated how ChatGPT-4o, using four prompting strategies (zero-shot, few-shot, Chain-of-Thought, and Generated Knowledge), classified whether pain or fatigue symptoms affected physical, social, or cognitive functioning.”

Response:

We appreciate this suggestion to improve clarity. We revised the Background and Methods sections of the **Abstract** to clarify the study's purpose and approaches. Specifically, the Background now states, "This study evaluates how two large language models (ChatGPT-4o and Llama-3.1) perform this task across different prompt engineering strategies," and the Methods state, "ChatGPT-4o and Llama-3.1 were evaluated using four prompting strategies: zero-shot, few-shot, step-by-step reasoning (Chain-of-Thought), and generated knowledge."

3. I somewhat feel the title could be tweaked regarding "classification of symptom narratives" I was expecting the paper to be around identifying whether symptoms were present, instead of classifying how symptoms are affecting a patient. Also, the authors should add "large" as they are not looking at language models generally here.

Response:

We appreciate this clarification. Our study focuses on classifying the functional impacts of pain and fatigue rather than detecting symptom presence alone. To avoid confusion, we revised the title to explicitly reflect functional impact classification and to specify the use of large language models: "Optimizing Prompting Strategies Improves Large Language Model Classification of Pain- and Fatigue-Related Functional Impact in Childhood Cancer Survivors."

Reviewer #2:

Thank you for the opportunity to read through this well-written and interesting paper. The manuscript explores the use of advanced NLP methods—particularly LLMs (ChatGPT)—for symptom interpretation in childhood cancer survivorship. The study applies zero-shot, few-shot, and Chain-of-Thought prompting approaches to model semantic representations of pain and fatigue. The integration of structured frameworks (PROMIS) and fairness considerations supports both methodological rigor and clinical sensitivity. Overall, the work is timely and relevant, offering novel insights into the intersection of AI, cancer care, and patient-reported outcomes. However, a few clarifications, definitional refinements, and structural adjustments are needed to strengthen the paper.

1. The manuscript introduces concepts such as bidirectional contextual representations of text, zero-shot prompting, few-shot prompting, and Chain-of-Thought prompting. While these are well described in the Methods section, they could be first presented in the Introduction to enhance readability through the paper as structured (Introduction>Results>Discussion>Methods).

Response:

We thank the reviewer for this helpful suggestion to improve readability. In the revised manuscript, we introduced key concepts related to prompting strategies in the **Introduction** section to establish a foundational understanding before these concepts are presented in the **Results** section, while retaining detailed methodological descriptions in the **Methods** section.

Specifically, we included concise definitions in the fourth paragraph of the **Introduction** section to read: "LLMs can be applied to classification tasks using prompting strategies that differ in complexity and information requirements.²³ Zero-shot prompting relies solely on task instructions; few-shot prompting incorporates a small set of labeled examples; Chain-of-Thought (COT) prompting elicits step-by-step reasoning; and Generated Knowledge (GK) prompting

draws on background knowledge before classification. Evaluating their performance in symptom analysis is essential for optimizing clinical LLM applications.”

2. Please ensure acronyms are defined on first use. For example, I think there is an issue with “Chain-of-Thought” and “Chain-of-Thought (COT)”.

Response:

We thank the reviewer for catching this inconsistency. We have reviewed the entire manuscript and ensured all acronyms are defined adequately at first use and used consistently thereafter.

3. Terms such as accuracy (0.740, 95% CI: 0.658–0.809), precision (0.712, 95% CI: 0.577–0.817), recall, and F1-score require clearer interpretation in the context of symptom classification. Specifically, what constitutes “good” precision for these models? Given the clinical implications of false positives versus false negatives, some discussion is warranted.

Response:

Thank you for raising this important point regarding clinical interpretation of model performance metrics. We have revised the first and second paragraphs of the **Discussion** section to explicitly contextualize accuracy, precision, recall, F1-score, and AUROC in the setting of symptom classification and survivorship care:

“This study evaluated four prompt-engineering strategies using ChatGPT-4o and Llama-3.1 to classify pain- and fatigue-related physical, social, and cognitive functional impact from symptom narratives. Across both domains, ChatGPT-4o with GK demonstrated the strongest and most consistent performance, achieving AUROC values of 0.73-0.78 with F1 scores of 0.50-0.69 for pain and AUROC values of 0.81-0.83 with F1 scores of 0.53-0.67 for fatigue, indicating stable discrimination across all three functional domains. COT prompting improved discrimination in selected domains, most notably for pain-related social functioning (AUROC up to 0.89), but showed greater variability than GK. In contrast, Llama-3.1 exhibited a recall-dominant profile, achieving near-perfect sensitivity for pain-related physical functioning (1.00) and high sensitivity for fatigue-related social and cognitive functioning (0.78-0.94), but with low precision (0.19-0.28) and modest F1 scores (0.32-0.42), resulting in a substantial false-positive burden. Zero-shot and few-shot prompting produced low and unstable performance across both models.

These performance patterns translate into distinct clinical use profiles for the two LLMs. ChatGPT-4o with GK, which maintained a more balanced precision–recall tradeoff and stable discrimination across functional domains, is better aligned with clinician-reviewed decision support and longitudinal symptom monitoring, where reliability and avoidance of unnecessary follow-up are critical. In contrast, Llama-3.1’s sensitivity-heavy, low-precision profile makes it more suitable for screening and surveillance applications, where minimizing missed impairments is prioritized, and downstream triage is expected. The poor and inconsistent performance of zero-shot and few-shot prompting across both models indicates that simple instruction-based prompting is insufficient for clinical-grade symptom classification. Importantly, neither model achieves the precision required for autonomous deployment, underscoring the need for clinician oversight and quality control in AI-assisted PRO interpretation.”

4. The manuscript also references area under the receiver operating characteristic curve

(AUROC) and area under the precision-recall curve (AUPRC). These are important but may require explanation for a mixed clinical/technical readership.

Response:

We appreciate this suggestion to enhance accessibility for clinician readers. We have added clear explanations in the Model Learning and Evaluation subsection under the **Methods** section, as follows: “AUROC quantifies a model’s ability to distinguish between positive and negative cases across all classification thresholds, ranging from 0.5 (no better than chance) to 1.0 (perfect discrimination), with values >0.7 indicating acceptable and >0.8 indicating good discrimination. AUPRC is particularly informative for imbalanced datasets, emphasizing true-positive detection while accounting for false positives, an important consideration when unnecessary follow-up can strain clinical resources. Metrics interpretation was informed according to clinical risk tolerance: 1) screening and surveillance prioritize high sensitivity (recall >0.80) to minimize missed clinically relevant cases (false negatives); 2) clinical decision support allows balanced missed cases and false-positive alarms (F1 0.60-0.75), consistent with routine practice; and 3) autonomous or unsupervised clinical decision-making requires very high precision (>0.90) to limit harmful false positives.”

5. It remains unclear why pain and fatigue were prioritized over other dimensions of symptom experience.

Response:

We appreciate the opportunity to clarify this point and have added clear explanations in the Data Collection and Annotation subsection under the **Methods** section, as follows: “For the present analysis, we focused on pain and fatigue, extracting 395 semantic units for pain and 424 for fatigue, each capturing symptom presence, intensity, or functional impact. This focus was driven by several considerations. First, pain and fatigue are among the most prevalent and burdensome symptoms in childhood cancer survivors, with our prior work reporting prevalence rates of 40% and 45%, respectively, among survivors aged 8-17.9 years.⁸ Second, both symptoms affect multiple functional domains (physical, social, cognitive), making them well-suited for evaluating the performance of prompting strategies.⁸ Third, pain and fatigue are well characterized within established PROMIS frameworks, providing strong conceptual foundations for annotation and validation.³⁵ Finally, our team previously developed BERT-based models for these domains,¹⁴ enabling direct comparison between traditional NLP and LLM approaches using identical gold-standard annotations.”

6. Relatedly, the terms “social attribute” and “cognitive attribute” are not common in clinical/symptom science vernacular. Alternative phrasing should be considered to align more closely with established clinical terminology.

Response:

We agree with this suggestion and have updated the terminology throughout the manuscript to align with established symptom science frameworks, specifically replacing “social attribute,” “cognitive attribute,” and “physical attribute” with “social functioning impact,” “cognitive functioning impact,” and “physical functioning impact,” respectively.

In addition, we have defined physical, social and cognitive functional impact in the Data Collection and Annotation subsection under the **Methods** section, as follows: “Guided by the PROMIS framework,^{37,38} meaning units were categorized into three functional impact types:

physical (mobility, physical activities, and bodily sensations), social (relationships, social activities, and peer interactions), and cognitive (concentration, memory, and mental clarity). These categories reflect how symptoms interfere with daily life rather than diagnostic labels.”

7. The sentence “The integration of structured frameworks, such as the Patient-Reported Outcomes Measurement Information System (PROMIS) scales, further supported the clinical relevance of COT-guided interpretation” is somewhat ambiguous. Please clarify the connection between PROMIS, clinical relevance, and the COT interpretations.

Response:

We agree that this sentence lacked clarity. We have revised and expanded this discussion to explain the connection explicitly in the third paragraph of the **Discussion** section, as follows: “The effects were enhanced by embedding content from the PROMIS pain and fatigue scales^{37,38} into the prompts, grounding the models’ reasoning in clinically validated definitions of functional impact. For instance, the statement “I was too tired to go out with my friends” could reflect physical fatigue or social disengagement. In this case, GK prompts anchored classifications using pre-defined functional domains, while COT prompts parsed contextual cues, such as activities and relationships, to reduce misclassification.”

8. The discussion would benefit from more explicit commentary on practical applications. For instance: How might clinicians incorporate AI-driven interpretation of survivor-reported text into routine practice? What safeguards, training, or decision-support frameworks would be required for integration into clinical workflows?

Response:

This is an excellent suggestion. We have added a new section on practical implementation considerations to the fifth paragraph of the **Discussion** section, as follows: “Translating these findings into clinical practice requires workflow integration, safety safeguards, and clinician training. With effective prompt engineering, LLMs can serve as practical alternatives to traditional NLP approaches for symptom analysis, enabling researchers and clinicians to extract clinically meaningful information from unstructured text without specialized modeling infrastructure. LLMs could analyze free-text symptom narratives from patient portals, telehealth visits, or follow-up questionnaires and, when integrated into EHR documentation systems, automatically generate structured symptom-related physical, social, and cognitive functional impact labels from AI-assisted clinical transcripts such as Epic DAX Copilot.⁴⁴ Additionally, real-time interpretation could flag functional impairments requiring timely intervention, such as cognitive difficulties reflected in learning-related activities and prompting referrals for psychosocial support. Embedding LLMs in this workflow could also generate targeted clarification prompts (e.g., duration, severity, interference) when narratives are ambiguous, thereby enriching documentation, reducing clinician burden, and supporting earlier intervention and personalized survivorship care.⁴⁵ Because these capabilities can be implemented using either closed-weight (e.g., ChatGPT-4o) or open-weight (e.g., Llama-3.1) platforms, careful consideration of their respective strengths and limitations is essential (see **Table 5** for a comparison). Accordingly, safe deployment requires mandatory clinician review, disclosure of model uncertainty, audit trails for quality and bias monitoring, and periodic revalidation against gold-standard assessments. Targeted clinician training is also essential to ensure appropriate interpretation and override of automated suggestions. These tools are intended to augment, not replace, clinical judgment, supporting scalable, patient-centered clinical care. Future

implementation research is warranted to examine clinician perspectives and the impact of LLM use on therapeutic alliance and patient-provider trust.”

9. The dataset consisted of 395 semantic units for pain and 424 for fatigue. The adequacy of this dataset size requires explicit discussion. While qualitative richness may be high, quantitative reliability for training/testing modern LLMs is less certain. Please provide justification or acknowledge limitations.

Response:

Thank you for this important methodological comment. We have added a discussion of dataset size implications to the **Limitations** section, as follows: “Finally, our dataset of 395 pain-related and 424 fatigue-related meaning units is modest in size, which may limit generalizability, subgroup fairness assessment, detection of subtle differences across prompting strategies, and representation of rare symptom patterns. However, the qualitative richness of the semi-structured interviews partially mitigates these constraints by capturing authentic survivor language and contextual nuance. Ongoing data collection and cross-institutional collaborations will expand sample size while preserving the depth needed for accurate, context-sensitive symptom classification.”

10. In addition, the sentence reporting these dataset counts seems misplaced in its current location. I recommend moving it to the Data Collection section for clarity.

Response:

We agree and have relocated this information to the Data Collection and Annotation subsection of the **Methods** section, as follows: “Semi-structured interviews with 30 survivors and their caregivers yielded rich narrative data across symptom domains and were audio-recorded and transcribed verbatim.^{35,36} For the present analysis, we focused on pain and fatigue, extracting 395 semantic units for pain and 424 for fatigue, each capturing symptom presence, intensity, or functional impact.”

COMMSMED-25-1565A

Optimizing Prompting Strategies Improves Large Language Model Classification of Pain- and Fatigue-Related Functional Impact in Childhood Cancer Survivors

Overall Response

We are pleased that two reviewers are satisfied with our responses and appreciate Reviewer 1's one additional comment. In the response letter below, this comment is shown in *italics*, and our response is presented in regular font. In the revised manuscript, all changes are indicated using underlined red text.

Reviewer #1's Comment

I would like to thank the authors for their thorough and thoughtful revisions. In particular, I appreciate the addition of the open-weight model comparison (Llama-3.1-8B-Instruct) alongside GPT-4o, and the inclusion of prior BERT results in the comparative performance tables. These additions substantially strengthen the manuscript by improving transparency, reproducibility, and generalizability. I also thank the authors for engaging with my concern regarding potential information leakage. In the rebuttal, they clearly explain that the dataset was split at the meaning-unit level within each interview/participant (70% learning and 30% testing), and they appropriately acknowledge that within-participant clustering may inflate performance estimates. I further appreciate the clarification that the LLM-based/prompting approaches were not "trained" in the traditional sense.

However, I do not believe this limitation is yet sufficiently acknowledged in the revised manuscript itself. Even without model training, the approach still involves prompt engineering and procedure development on the learning subset; therefore, a within-participant split may still yield optimistic performance estimates due to lexical/stylistic dependence between learning and test meaning units from the same participant. This also limits conclusions about generalization to unseen patients. I therefore request that the authors explicitly add this point to the Limitations section and note that future work should assess robustness using participant-level separation (e.g., patient-level train/test split). Assuming this is added, I have no further comments.

Response

We appreciate the reviewer's careful reading and agree that this limitation warrants more explicit acknowledgment. Although the large language models were not trained or fine-tuned on study data, the analytic workflow, including prompt engineering and evaluation procedures, was developed using a learning subset of meaning units. Because the train-test split was conducted at the meaning-unit level within participants, residual lexical or stylistic similarity across symptom statements from the same individual may lead to optimistic performance estimates and constrain inference about generalizability to unseen patients.

To address this concern, we have added a statement to the Limitations section, which now reads: "Although the large language models were not trained or fine-tuned on study data, prompt engineering and evaluation procedures were developed using a learning subset of meaning units. Because training and testing were conducted at the meaning-unit level within participants, residual within-participant dependence may inflate performance estimates. Consequently, the findings primarily reflect classification performance at the symptom-statement level rather than robust generalization to unseen patients. Future studies should assess robustness using participant-level separation (e.g., patient-level train/test splits) in larger, multi-institutional datasets."